Exp No:5 Data Collection and Initial Exploration

Aim:

To collect, load, and perform initial exploration of the diabetes dataset.

Materials Needed:

Diabetes dataset (available from sources such as UCI Machine Learning Repository

or Kaggle)

Python with libraries: pandas, numpy, matplotlib, seaborn

Procedure and Code:

1. Data Collection:

```
db = pd.read_csv("diabetes_dataset_50.csv")
```

```
db.head()
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

2. Initial Exploration:

```
print(db.info)
```

```
print(db.describe)
```

```
import matplotlib.pyplot as plt
```

```
import seaborn
```

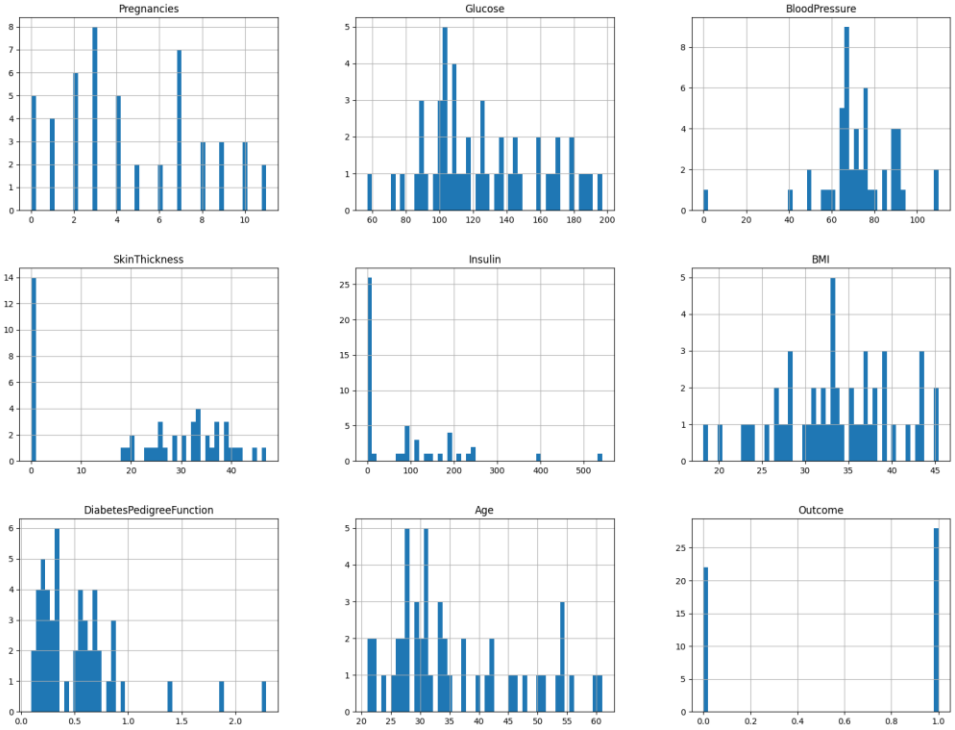
```
db.hist(bins=50,figsize=(20,15))
```

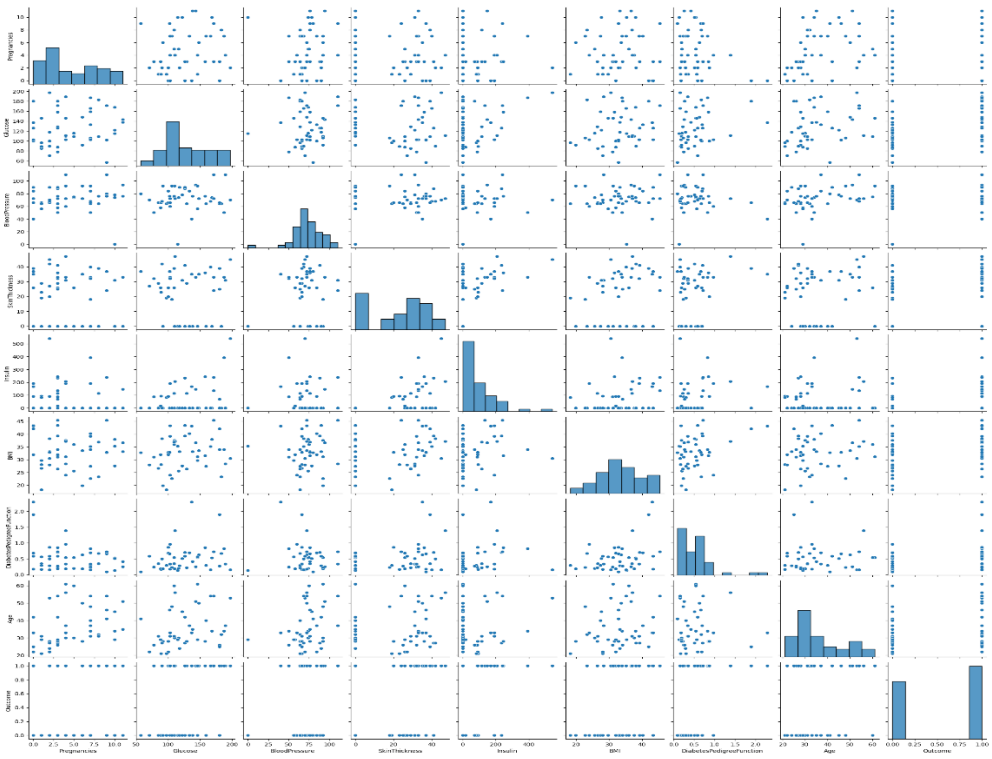
```
plt.show()
```

```
import seaborn as sns
```

```
sns.pairplot(db)
```

```
plt.show()
```





Result:

Thus the diabetes dataset collected, loaded and perform initial exploration of the diabetes dataset